

The finite density-transfer theorem

Claude, with Daniel Murfet

2026-09-07

Abstract

An abstract, finite, quantitative form of “a power-log expansion of the state density transfers to a power-log asymptotic expansion of the partition function”. For $Z(N) = \int_0^R e^{-\beta(Nr)^2} \rho(r, Nr) dr$ with a two-variable density satisfying $\rho(r, s) = \sum_i r^{\alpha_i-1} \log^{j_i}(1/r) c_i(s) + E(r, s)$, $|E| \leq r^{a-1}(1 + |\log r|)^J H(s)$, $\alpha_i < a$, $j_i \leq J$, and finite weighted moments,

$$\left| Z(N) - \sum_i N^{-\alpha_i} \sum_{m \leq j_i} \binom{j_i}{m} (\log N)^m (-1)^{j_i-m} M_{i, j_i-m} \right| \leq N^{-a} (1 + \log N)^J \left(M_H + \sum_i R^{\alpha_i-a} T_i \right) \quad (N \geq 1),$$

with $M_{i,\ell} = \int_0^\infty s^{\alpha_i-1} (\log s)^\ell e^{-\beta s^2} c_i(s) ds$. This is Astra’s target 7 in the form recommended in the second consult. Zero sorry/axiom.

1 The things formalised

```
noncomputable def transferZ ( : → → ) ( R N : ) : :=
  r in Ioc (0 : ) R, Real.exp ( - * (N * r) ^ 2 ) * r (N * r)
noncomputable def logMoment ( : ) ( : ) (c : → ) : :=
  s in Ioi (0 : ), s ^ ( - 1 ) * Real.log s ^ * (Real.exp ( - * s ^ 2 ) * c s)
noncomputable def transferTerm ( : ) (j : ) (c : → ) (N : ) : :=
  N ^ ( - ) * m Finset.range (j + 1),
  Real.log N ^ m * (-1) ^ (j - m) * (j.choose m : ) * logMoment (j - m) c
structure DensityExpansion ( : → ) (j : → ) (c : → → ) (a R : ) (J : ) (H)
  -- R_pos, _lt, j_le, measurability, H 0, and
  rem : r Ioc 0 R, s, 0 < s →
    | r s - i, r ^ ( i - 1 ) * Real.log (1 / r) ^ j i * c i s|
    r ^ (a - 1) * (1 + |Real.log r|) ^ J * H s
theorem densityTransfer_bound (hD : DensityExpansion j c a R J H) ( : )
  (hMc : i, IntegrableOn (s ^ ( i - 1 ) (1 + |log s|) ^ (j i) e ^ {- s ^ 2} | c i s |) (Ioi 0))
  (hMt : i, IntegrableOn (s ^ (a - 1) (1 + |log s|) ^ (j i) e ^ {- s ^ 2} | c i s |) (Ioi 0))
  (hMH : IntegrableOn (s ^ (a - 1) (1 + |log s|) ^ J e ^ {- s ^ 2} H s) (Ioi 0))
  (N : ) (hN : 1 N) :
  |transferZ R N - i, transferTerm ( i) (j i) (c i) N|
  N ^ (-a) * (1 + Real.log N) ^ J
  * (envMoment a J H + i, R ^ ( i - a ) * tailMoment a (j i) (c i))
```

2 Proof strategy

Three independent mechanisms, each a lemma. Scaling: $\int_0^R F(Nr) dr = N^{-1} \int_0^{NR} F$ (`intervalIntegral.integral_comp_mul_left`), and integrability transports the same way (`IntervalIntegrable.comp_mul_left`). Binomial expansion: $(\log N - \log s)^j = \sum_m \binom{j}{m} (\log N)^m (-\log s)^{j-m}$ (`add_pow`), so the truncated term is a combination of truncated log moments. Tail replacement: on $s \geq NR$, $s^{\alpha-1} \leq (NR)^{\alpha-a} s^{a-1}$ and $|\log s|^\ell \leq (1 + |\log s|)^j$, whence the tail costs $N^{-a} R^{\alpha-a} T$ times $\sum_m \binom{j}{m} (\log N)^m = (1 + \log N)^j \leq (1 + \log N)^J$. The remainder is dominated pointwise after scaling via $1 + |\log s - \log N| \leq (1 + \log N)(1 + |\log s|)$.

3 Roadblocks and resolutions

- `IntervalIntegrable.comp_mul_left` takes the scale as a named implicit $((c := N))$ after two `autoParams`.
- `sub_eq_add_neg` rewrites the first subtraction it finds (the exponent $\alpha - 1$), not the intended $\log N - \log s$; rewrite the specific term with `show`.
- A helper name (`abs_log_pow_le`) clashed with an existing declaration in another file of the namespace; the clash surfaces only in the full `lake` build of `Laplace.lean`. **Gate lesson:** a shell pipeline that greps “error” returns success; gate on the build’s exit code, not on the grep.
- `Measurable.abs` does not exist; use `continuous_abs.measurable.comp`.

4 Outcome

The transfer theorem separates analytic estimates from the future combinatorial construction of the density expansion. Next: the parameterised first/second-order Taylor kernel lemma for $\eta(u, v)e^{\beta s \xi(u, v)}$ and the $d = 2$ anchored-subtraction density formula, whose output plugs into `densityTransfer_bound` to give the first complete nonconstant- ξ, η log polynomial.